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Big Data & AI Project

COVID-19 Risk Classification Using PySpark

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Abstract

The COVID-19 pandemic emphasised the importance of scalable, data-driven decision-support systems for public health authorities. This project presents a big data classification framework for predicting high-risk COVID-19 scenarios at the province-day level using heterogeneous datasets. Apache Spark and PySpark ML were employed to process large-scale data, perform exploratory data analysis, apply feature engineering, and train multiple classification models. Five baseline models and their optimised counterparts were evaluated using robust metrics including F1-score, AUROC, and average precision-recall. Experimental results demonstrate that Gradient Boosted Trees outperform other approaches, confirming their suitability for structured epidemiological data.

1 Introduction

The COVID-19 pandemic posed unprecedented challenges to healthcare systems worldwide. Rapid transmission rates, regional outbreaks, and limited medical resources made early risk identification a critical task. While dashboards and descriptive statistics provide situational awareness, they lack predictive capabilities required for proactive intervention.

This project addresses this gap by formulating COVID-19 risk prediction as a binary classification problem. Each province on a given day is classified as either *low-risk* or *high-risk* based on confirmed case intensity. By integrating demographic, geographical, behavioural, and temporal indicators, the proposed system aims to support data-driven public health decisions.

Big data technologies are essential due to the volume, variety, and velocity of COVID-19 datasets. Apache Spark was chosen for its distributed processing capabilities, fault tolerance, and native support for machine learning pipelines.

2 Problem Statement and Objectives

2.1 Problem Statement

The goal of this project is to predict whether a province on a given day represents a high COVID-19 risk scenario. The challenge lies in combining heterogeneous data sources while avoiding data leakage and overly optimistic performance.

2.2 Objectives

- Analyse multi-source COVID-19 datasets using PySpark
- Perform extensive exploratory data analysis (EDA)
- Engineer and select relevant features
- Train and compare multiple classification algorithms
- Optimise models using cross-validation
- Select the best-performing model using robust metrics

3 System Architecture

To ensure scalability and real-world applicability, multiple deployment architectures were considered.

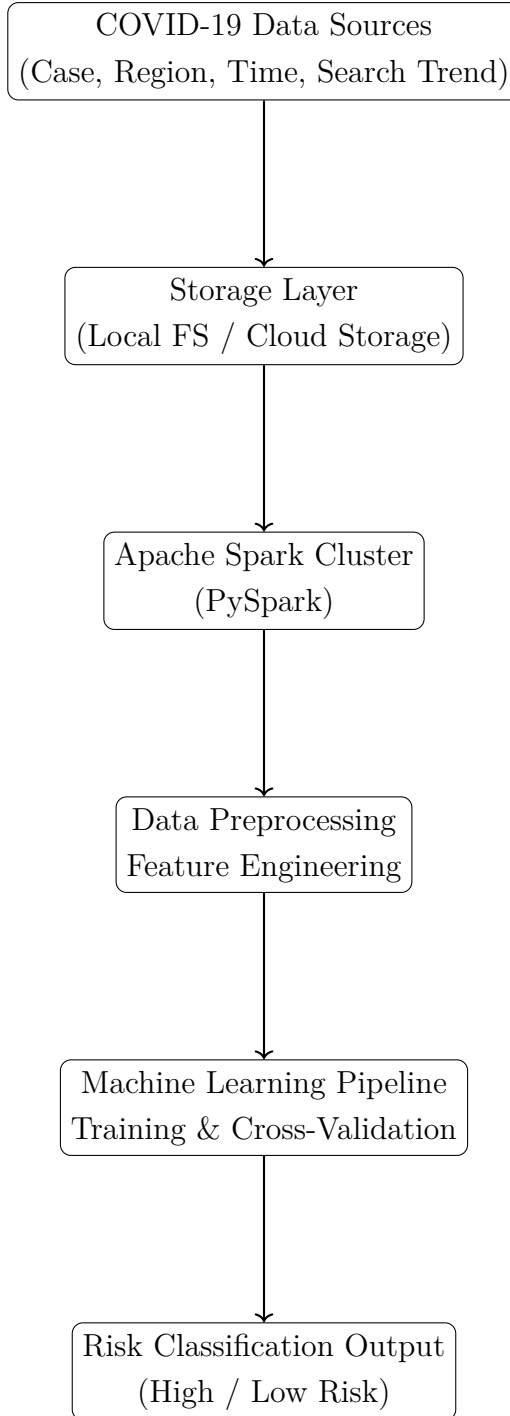


Figure 1: Proposed System Architecture for COVID-19 Risk Classification

3.1 Databricks Batch Architecture

Raw datasets are stored in cloud object storage, while Spark jobs running on Databricks perform preprocessing, training, and batch inference. This approach offers minimal operational overhead and strong scalability.

3.2 AWS EMR with S3

Spark clusters on AWS EMR process data stored in Amazon S3. This architecture offers flexibility in cluster configuration and cost optimisation.

3.3 Azure Synapse Analytics

Azure Synapse integrates Spark pools with cloud storage and analytics services, enabling seamless execution of large-scale machine learning workloads.

3.4 On-Premise Spark Cluster

For strict data governance requirements, an on-premise Spark cluster can be used, ensuring full control over sensitive data.

4 Datasets Description

The project integrates multiple datasets, including:

- Province-level COVID-19 case statistics
- Demographic and geographical information
- Behavioural search trends
- Gender-based confirmed case counts

All datasets were joined at the province-day level to construct a unified analytical dataset.

5 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand data distributions, relationships, and anomalies.

5.1 Key Performance Indicators

- Average confirmed cases per province
- Maximum daily confirmed cases
- Elderly population ratio distribution
- Nursing home density

- Male vs female confirmed case ratio
- COVID-related search trend intensity
- High-risk vs low-risk class distribution

5.2 Visual Analysis

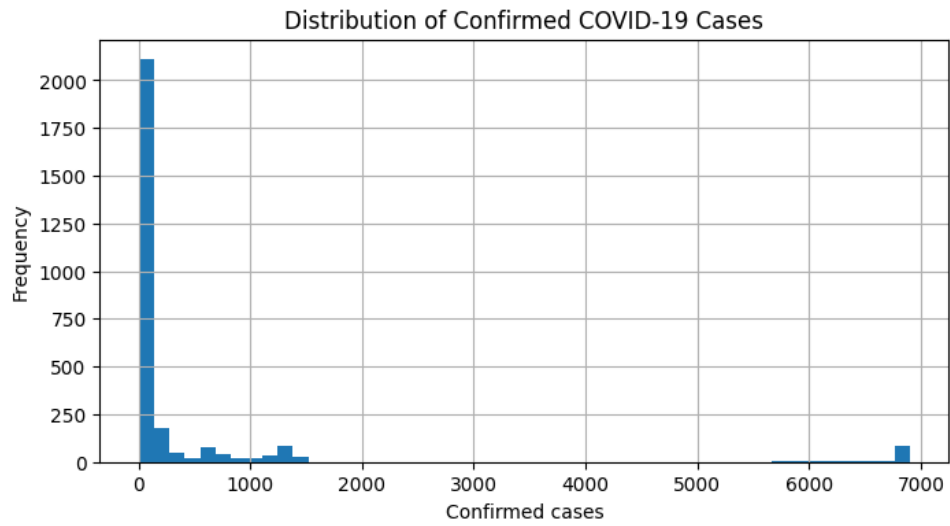


Figure 2: Distribution of Confirmed COVID-19 Cases

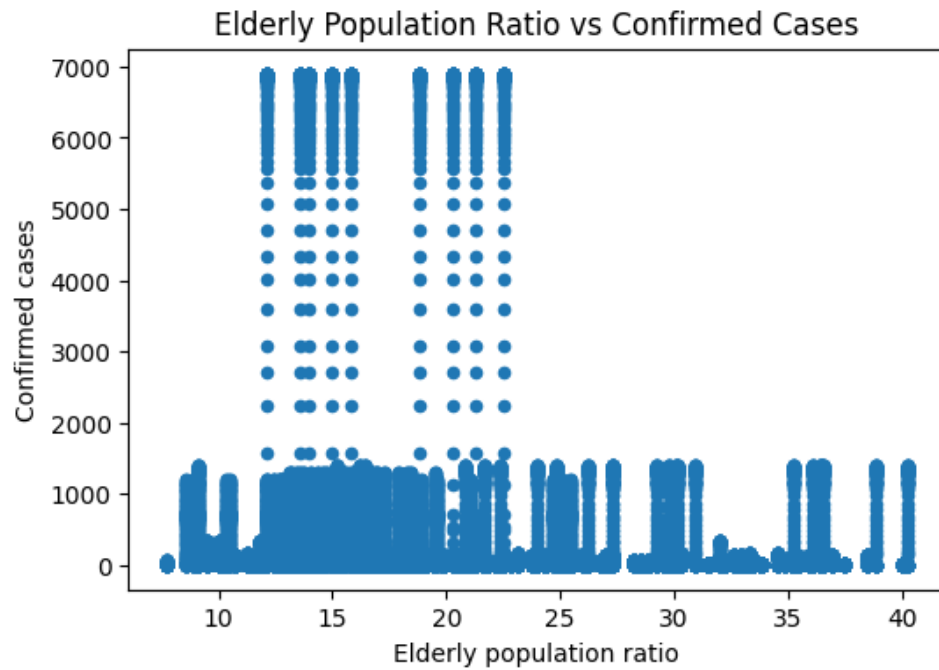


Figure 3: Elderly Population Ratio vs Confirmed Cases

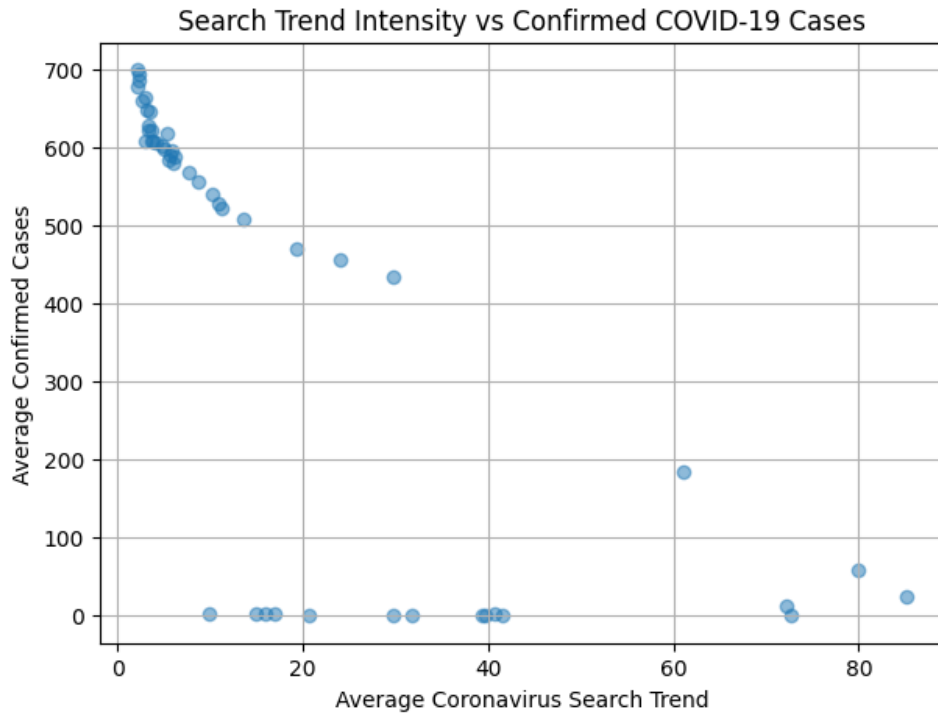


Figure 4: Search Trend Intensity vs Confirmed Cases

EDA revealed strong correlations between confirmed cases and both demographic and behavioural indicators.

6 Data Preprocessing

Data preprocessing included:

- Handling missing values
- Encoding categorical variables
- Scaling numerical features
- Preventing data leakage

A percentile-based threshold was applied to confirmed cases to generate a binary target variable.

7 Feature Selection and Importance

Feature selection was performed manually based on domain knowledge and insights gained during exploratory data analysis. The objective was to reduce dimensionality, improve model generalisation, and avoid proxy leakage from highly correlated or outcome-related

variables. Both numerical and categorical features were explicitly selected and incorporated into the machine learning pipeline.

To validate the relevance of the selected features and to improve model interpretability, a model-based feature importance analysis was conducted using a Random Forest classifier trained on the engineered feature vector. Random Forest feature importance provides an estimate of each feature’s contribution to the classification decision by measuring its impact on impurity reduction across the ensemble.

The analysis highlighted the strong influence of demographic and behavioural indicators, particularly:

- Elderly population ratio
- Nursing home count
- COVID-related online search trends

These results are consistent with epidemiological understanding, as regions with higher elderly populations and greater institutional care density are more vulnerable to severe outbreaks, while increased search activity often reflects heightened public awareness during periods of rising infections. Feature importance was used for validation and interpretability rather than automatic feature elimination, ensuring stability and robustness of the modelling pipeline.

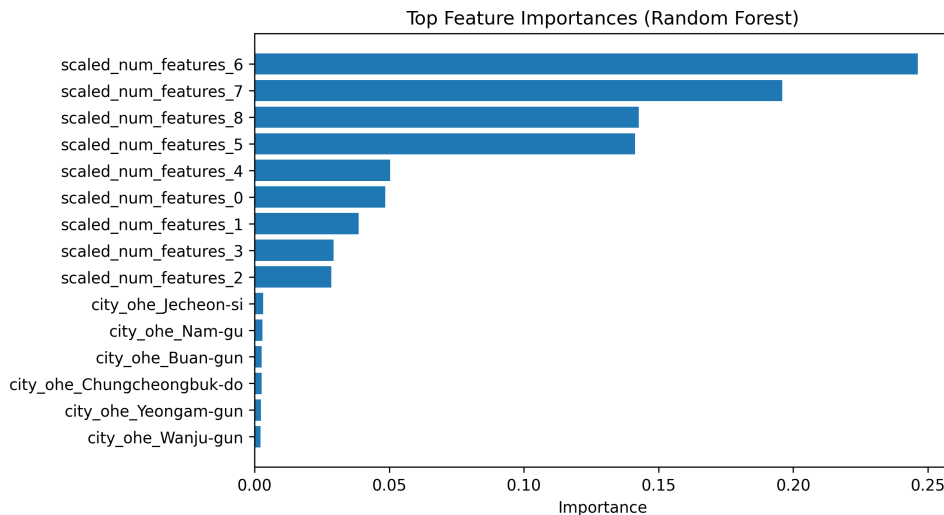


Figure 5: Top Feature Importances Computed Using a Random Forest Classifier

8 Machine Learning Models

Five baseline models were evaluated:

- Logistic Regression

- Decision Tree
- Random Forest
- Gradient Boosted Trees
- Neural Network (Multilayer Perceptron)

9 Model Optimisation

Hyperparameter optimisation was performed using grid search with cross-validation, improving robustness and generalisation.

10 Results and Evaluation

10.1 Baseline Model Performance

Model	Accuracy	F1-score
Logistic Regression	0.9654	0.9654
Decision Tree	0.8380	0.8371
Random Forest	0.8202	0.8171
Gradient Boosted Trees	0.9914	0.9914
Neural Network	0.7115	0.6918

Table 1: Baseline Model Performance

10.2 Optimised Model Performance

Model	Accuracy	F1-score
Logistic Regression (CV)	0.9501	0.9501
Decision Tree (CV)	0.9910	0.9910
Random Forest (CV)	0.9059	0.9057
Gradient Boosted Trees (CV)	0.9919	0.9919
Neural Network (CV)	0.8098	0.8081

Table 2: Optimised Model Performance

10.3 Additional Metrics

The Gradient Boosted Trees model achieved high AUROC and average precision-recall scores, confirming strong discriminative capability under class imbalance.

11 Final Model Selection

Gradient Boosted Trees with cross-validation was selected as the final model due to its superior performance, robustness, and interpretability.

12 Conclusion

This project demonstrates the effectiveness of big data analytics and ensemble learning for COVID-19 risk prediction. Through disciplined preprocessing, fair evaluation, and systematic optimisation, Gradient Boosted Trees emerged as the most suitable model. The proposed framework is scalable, interpretable, and applicable to real-world public health decision support.

13 References

- Apache Spark Documentation
- PySpark ML Guide
- COVID-19 Open Data Sources